

Q1

a.

	red	blue	green	grey
red	1	0	0	0
blue	0	1	0	0
blue	0	1	0	0
red	1	0	0	0
green	0	0	1	0
grey	0	0	0	1

← one-hot matrix

b. $\hat{P}_{red} = \frac{2}{6} = \frac{1}{3}$, $\hat{P}_{blue} = \frac{2}{6} = \frac{1}{3}$, $\hat{P}_{green} = \frac{1}{6}$, $\hat{P}_{grey} = \frac{1}{6}$

c. $\hat{F}_n(1) = \frac{1}{5} \sum_{i=1}^5 I\{x_i \leq 1\} = \frac{1}{5} = 0.2$

$\hat{F}_n(3) = \frac{1}{5} \sum_{i=1}^5 I\{x_i \leq 3\} = \frac{3}{5} = 0.6$

$\hat{F}_n(6) = \frac{1}{5} \sum_{i=1}^5 I\{x_i \leq 6\} = \frac{4}{5} = 0.8$

$\hat{F}_n(11) = \frac{1}{5} \sum_{i=1}^5 I\{x_i \leq 11\} = 1$

$$\hat{F}_n(x) = \begin{cases} 0, & x < 1 \\ 0.2, & 1 \leq x < 3 \\ 0.6, & 3 \leq x < 6 \\ 0.8, & 6 \leq x < 11 \\ 1, & x \geq 11 \end{cases}$$



Q2

a. $F_Y(y) = P(Y \leq y)$
 $= P(a + bX \leq y)$
 $= P(X \leq \frac{y-a}{b})$
 $= F_X(\frac{y-a}{b})$

$f_Y(y) = F'_Y(y)$
 $= F'_X(\frac{y-a}{b})$
 $= \frac{1}{b} f_X(\frac{y-a}{b})$

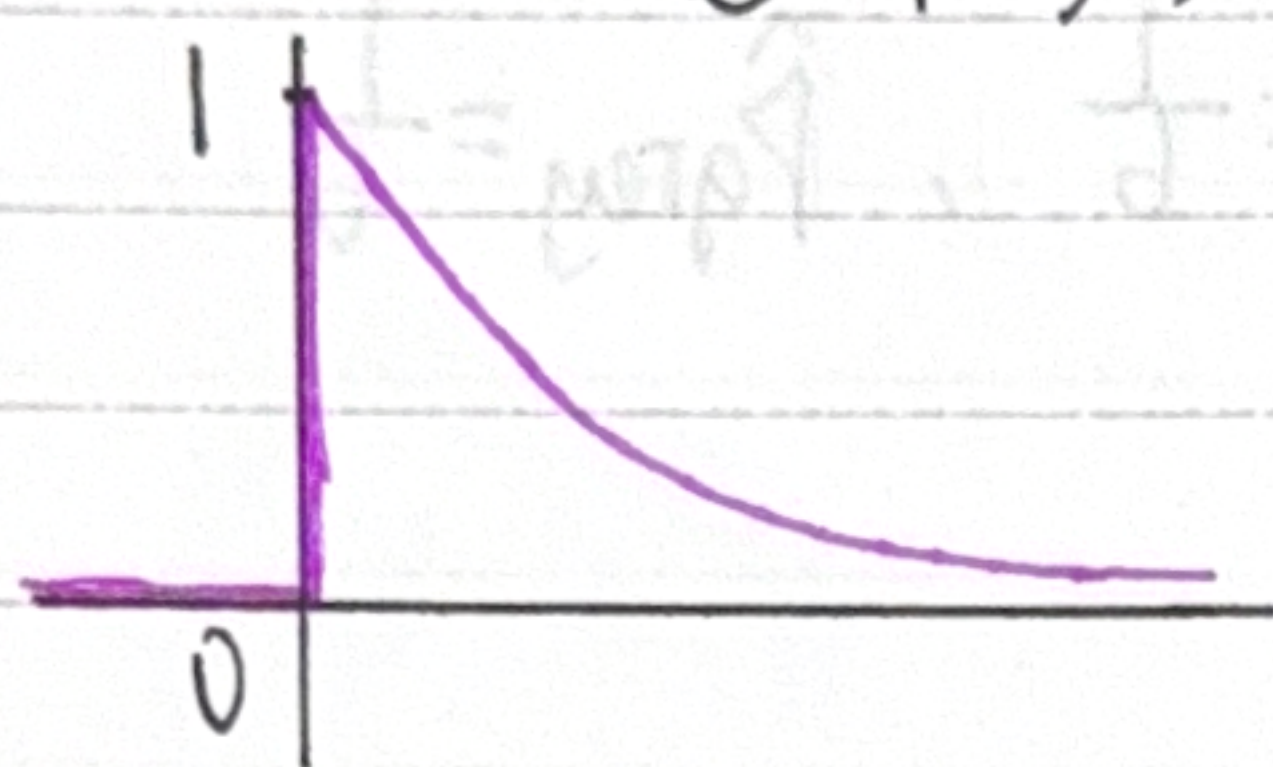
$$b. \bar{X} = \frac{1+3+4}{3} = \frac{8}{3}$$

$$c. \bar{X}^2 = \frac{1^2+3^2+4^2}{3} = \frac{1+9+16}{3} = \frac{26}{3}$$

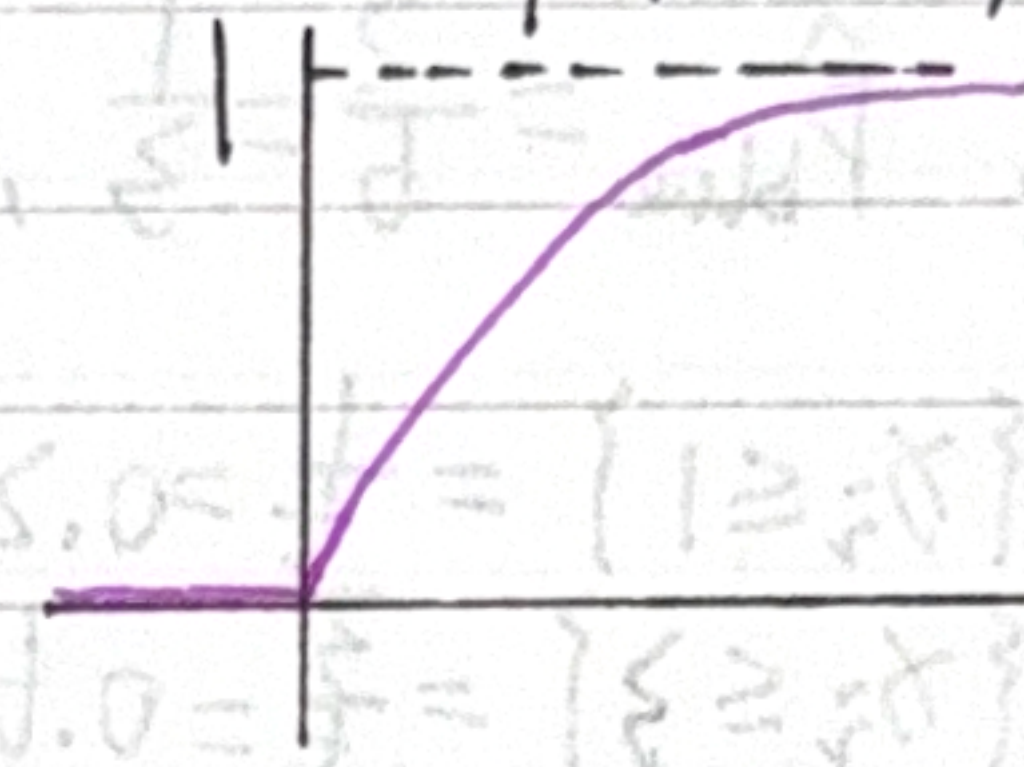
$$d. (\bar{X})^2 = \left(\frac{8}{3}\right)^2 = \frac{64}{9} \neq \bar{X}^2$$

Q3

$$a. f_X(x) = \begin{cases} 0, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases}$$



$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$



$$b. P(X \geq 3) = 1 - F_X(3) \\ = 1 - (1 - e^{-3}) \\ = e^{-3}$$

$$P(X \leq 3.5) = F_X(3.5) = 1 - e^{-3.5}$$

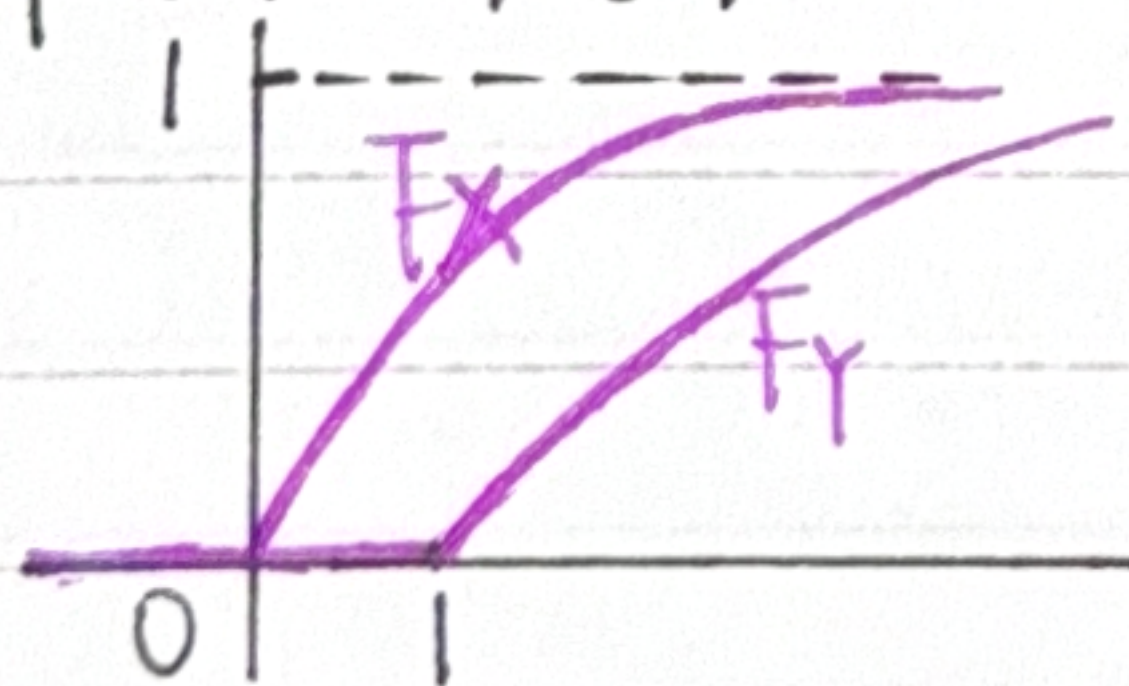
$$c. F_Y(y) = P[Y \leq y] \\ = P[1 + 3X \leq y] \\ = P[X \leq \frac{y-1}{3}] \\ = F_X\left[\frac{y-1}{3}\right]$$

$$X \geq 0, Y = 1 + 3X \geq 1$$

$$F_Y(y) = \begin{cases} 0, & y < 1 \\ 1 - e^{-\frac{y-1}{3}}, & y \geq 1 \end{cases}$$

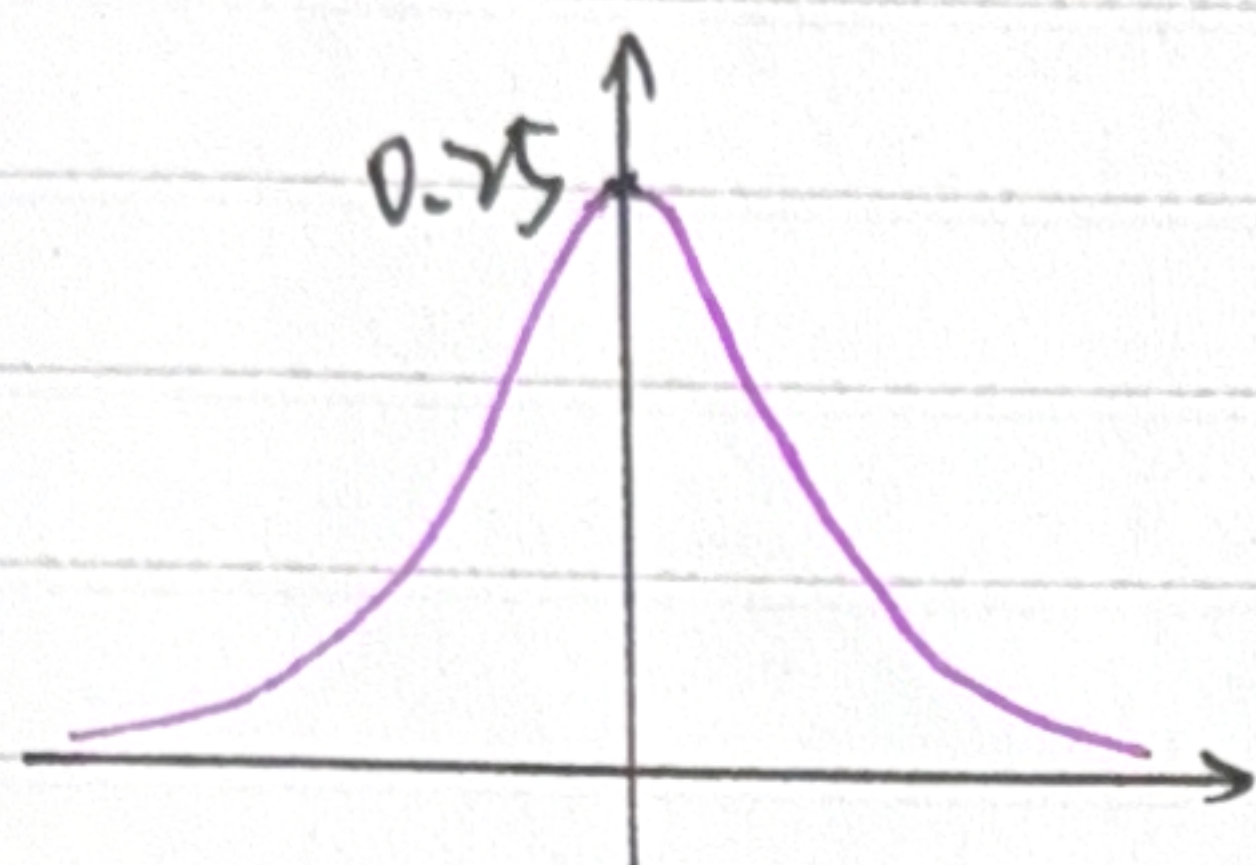
$$f_Y(y) = \begin{cases} 0, & y < 1 \\ \frac{1}{3} e^{-\frac{y-1}{3}}, & y \geq 1 \end{cases}$$

$$F_Y(10) = P[Y < 10] = 1 - e^{-\frac{10-1}{3}} = 1 - e^{-3}$$

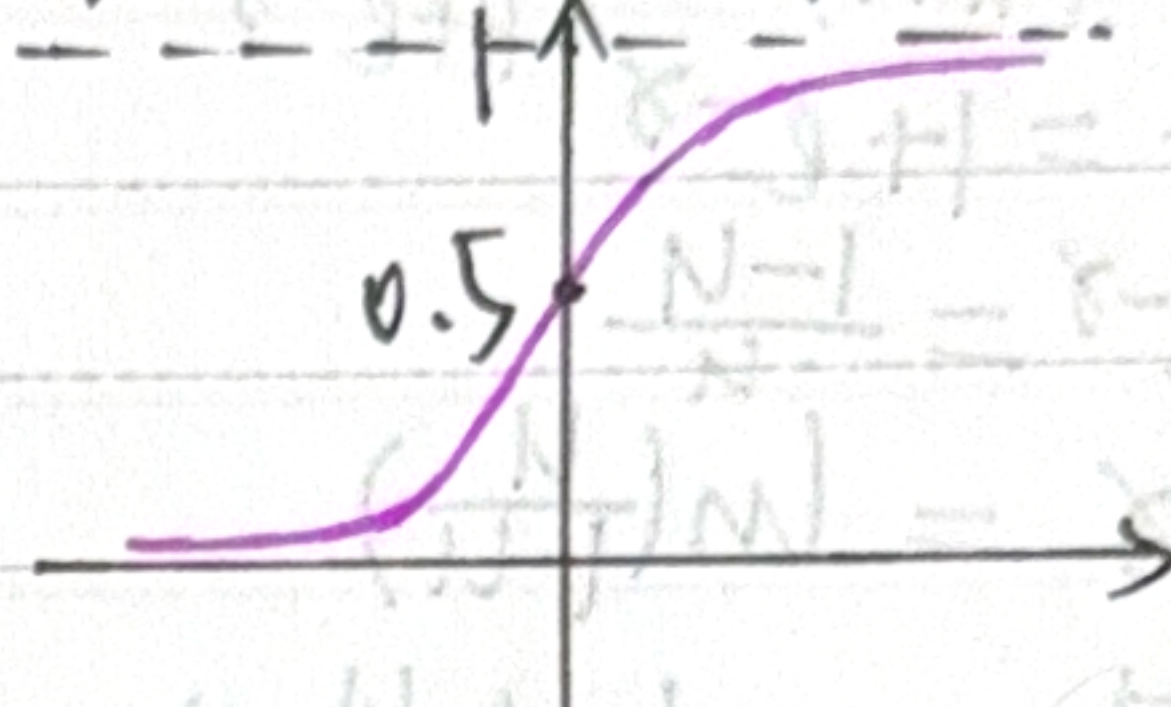


Q4

a. $f(x) = F'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$
 $x=0, f(x) = 0.25$



$F(x) = \frac{1}{1+e^{-x}}$
 $x \rightarrow -\infty, F(x) \rightarrow 0$
 $x \rightarrow \infty, F(x) \rightarrow 1$
 $x=0, F(x) = 0.5$

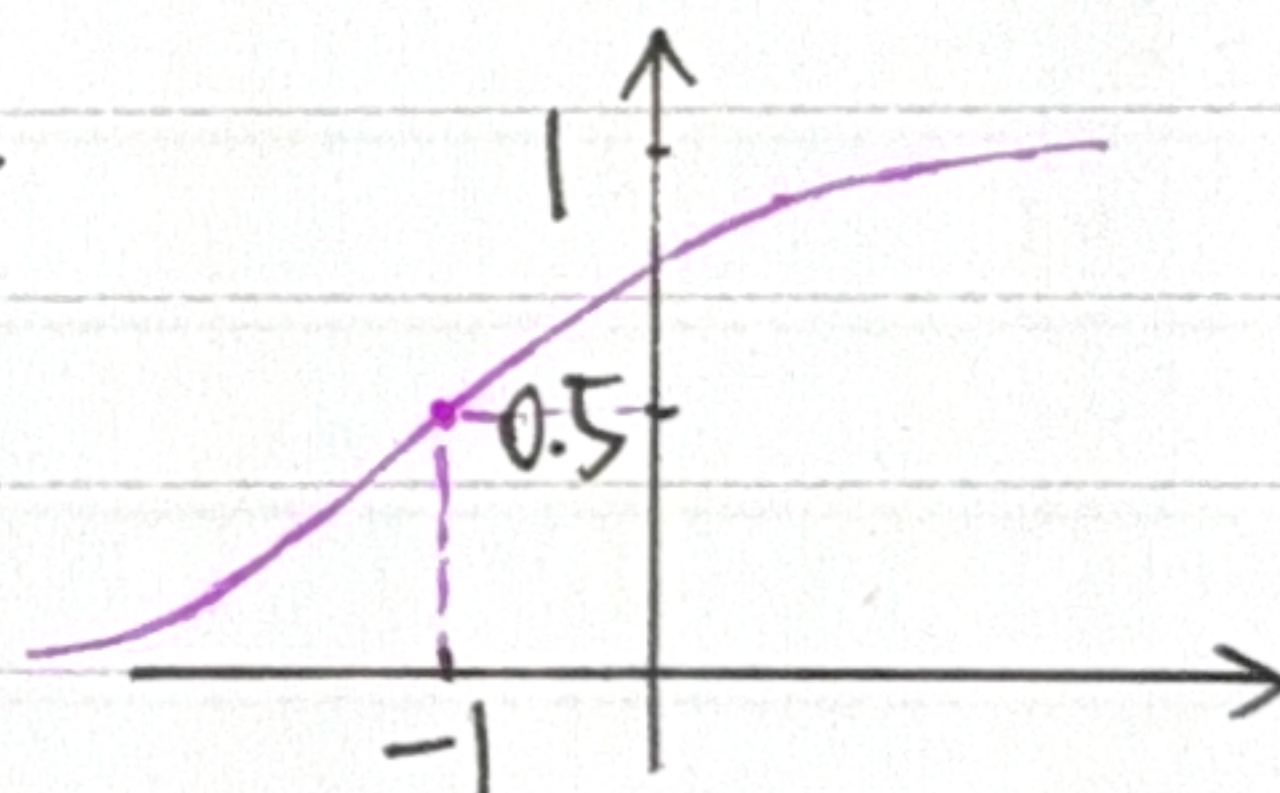


b. $P(X \geq 0.8) = 1 - P(X < 0.8)$
 $= 1 - F(0.8)$
 $= 1 - \frac{1}{1+e^{-0.8}} \approx 0.31$

$P(X \leq 0.3) = F(0.3) = \frac{1}{1+e^{-0.3}} \approx 0.57$

c. $F_Y(y) = P(Y \leq y) = P(2X-1 \leq y) = P(X \leq \frac{y+1}{2})$
 $= F_X(\frac{y+1}{2}) = \frac{1}{1+e^{-\frac{y+1}{2}}}$ $F_Y(-1) = 0.5$

$P(Y < 0) = F_Y(0) = \frac{1}{1+e^{-\frac{1}{2}}} \approx 0.62$



Q5

a. $F_X(x) = 1 - e^{-x}$, $F_X(m) = \frac{1}{2}$
 $1 - e^{-m} = \frac{1}{2}$
 $m = \ln 2$

b. $F_X(x) = \frac{1}{1+e^{-x}}$, $F_X(m) = \frac{1}{2}$
 $\frac{1}{1+e^{-m}} = \frac{1}{2}$
 $e^{-m} = 1$
 $m = 0$

$$c. u = F_X(x) = 1 - e^{-x}$$

$$e^{-x} = 1 - u$$

$$x = -\ln(1-u)$$

$$F^{-1}(u) = -\ln(1-u), \quad 0 < u < 1$$

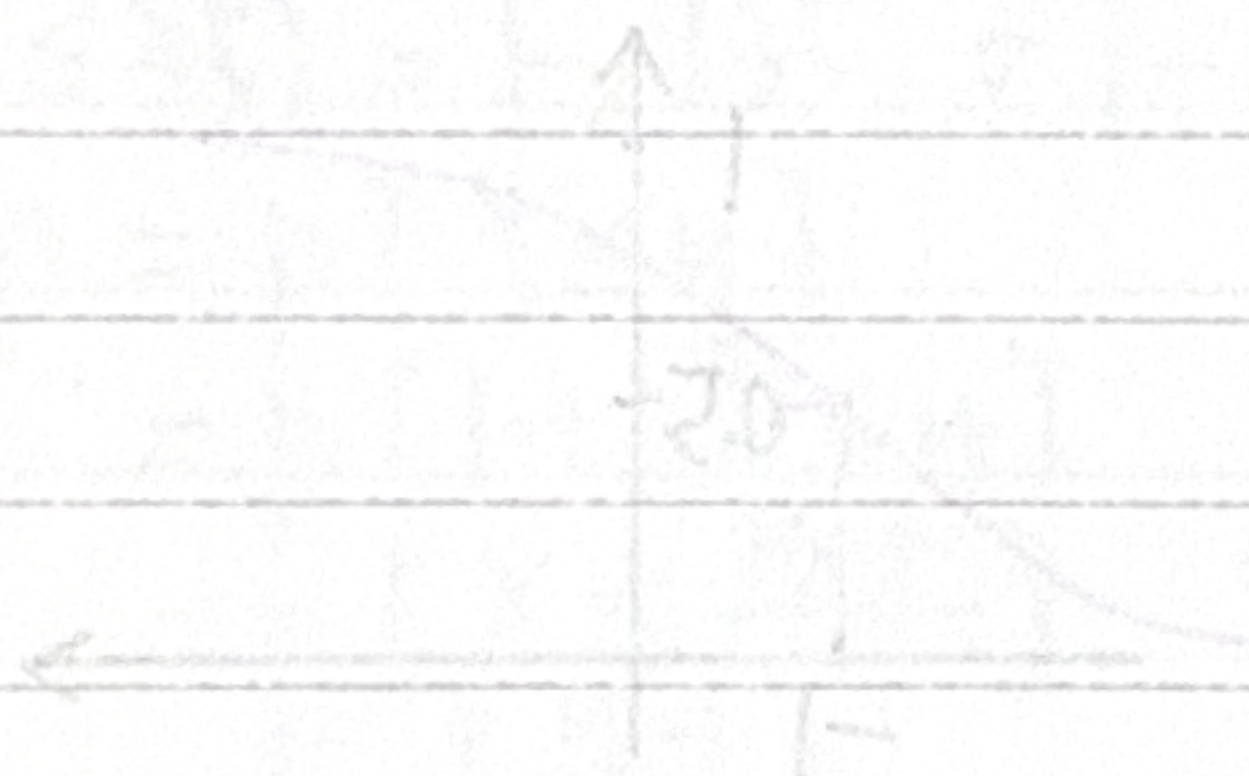
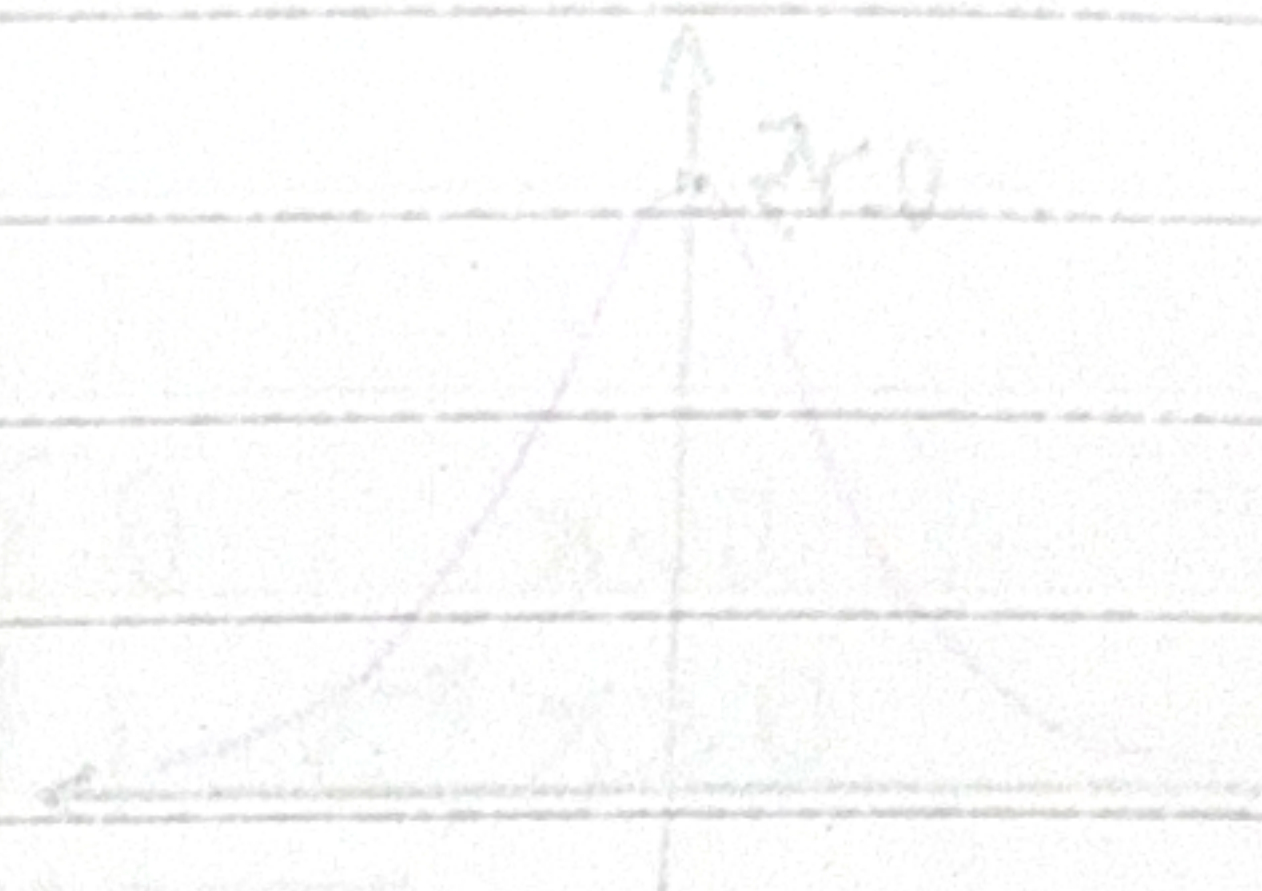
$$d. u = F_X(x) = \frac{1}{1+e^{-x}}$$

$$\frac{1}{u} = 1 + e^{-x}$$

$$e^{-x} = \frac{1-u}{u}$$

$$x = \ln\left(\frac{u}{1-u}\right)$$

$$F^{-1}(u) = \ln\left(\frac{u}{1-u}\right), \quad 0 < u < 1$$



$$\frac{1}{2} = F(x) = 1 - e^{-x} \Rightarrow \frac{1}{2} = 1 - e^{-x} \Rightarrow e^{-x} = \frac{1}{2} \Rightarrow -x = \ln\left(\frac{1}{2}\right) \Rightarrow x = \ln(2)$$

$$\frac{1}{2} = F(x)$$

$$m = m$$

$$\frac{1}{2} = F(x) = \frac{1}{1+e^{-x}} \Rightarrow \frac{1}{2} = \frac{1}{1+e^{-x}} \Rightarrow 1+e^{-x} = 2 \Rightarrow e^{-x} = 1 \Rightarrow -x = \ln(1) \Rightarrow x = 0$$

$$\frac{1}{2} = F(x)$$